Econ 381: Introduction to Econometrics
Handout 7: Multiple Regression

Muyang Ren

Department of Economics
University of Tennessee, Knoxville

Where Handout 6 left us

For the California schools, we estimated

$$\widehat{\text{TestScore}} = 698.93 - 2.28\,STR, \qquad \text{SE}(\widehat{\beta}_1) = 0.52.$$

The 95% confidence interval is approximately $[-3.30, -1.26]$.

The superintendent's policy question

Would reducing the student–teacher ratio actually improve test scores?

A confidence interval describes **sampling uncertainty**. It cannot tell us whether the slope also reflects **other** differences across districts.

A precisely estimated relation can still be a misleading causal effect.

Roadmap

We extend the same class size example one step at a time:

1. **6.1: Omitted variable bias**.
   Why can the simple regression analysis mislead us?

2. **6.2–6.3: Multiple regression and OLS**.
   Compare districts while holding other characteristics fixed.

3. **6.4: Measures of fit**.
   What improves when we add a regressor?

4. **6.5–6.6: Assumptions and sampling uncertainty**.
   Extend the tools from Handouts 5 and 6.

5. **6.7: Multicollinearity**.
   Can the data separate the regressors' roles?

6. **6.8: Controls and causality**.
   What makes a comparison credible?

Section 6.1

Omitted variable bias

What is hidden in the regression error?

Our simple model is

$$\text{TestScore}_i = \beta_0 + \beta_1 STR_i + u_i.$$

The error u_i collects determinants of achievement omitted from this equation:

- students' English proficiency and family background;
- teacher quality and other school resources.

Recall assumption A1 from Handout 5:

$$E(u_i \mid STR_i) = 0.$$

At each class size, omitted factors must average to zero.

If larger classes systematically have less favorable omitted factors, the estimated slope mixes those differences with the role of class size.

Two conditions for omitted variable bias

An omitted variable W creates bias in the simple slope when:

1. W **helps determine** Y: its effect is part of u.
2. W **is correlated with** X: its contribution to Y varies systematically with X.

Omitted factor	Relationship with scores	Relationship with STR
English learner share	Higher share may lower scores	Higher share is associated with larger STR
Teacher quality	Better teachers may raise scores	Suppose better teachers work in smaller classes

Both examples can make the simple slope **too negative**.

Class size and English learner share

Split the California districts by their percentage of English learners (PctEL):

	$PctEL < 10\%$	$PctEL \geq 10\%$
Number of districts	228	192
Average *STR*	19.37	19.96
Average test score	663.48	643.08

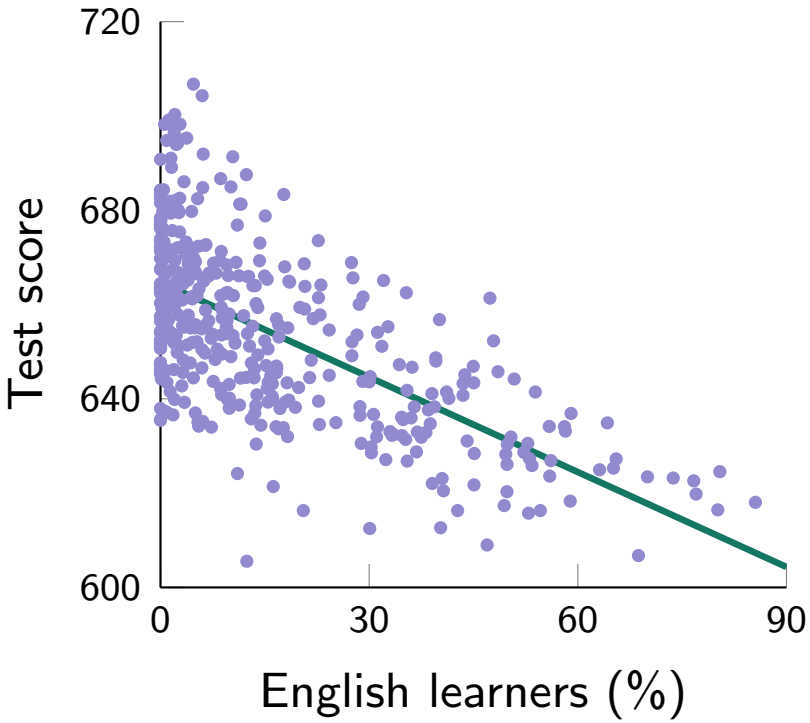
The second group has both larger classes and lower average scores.

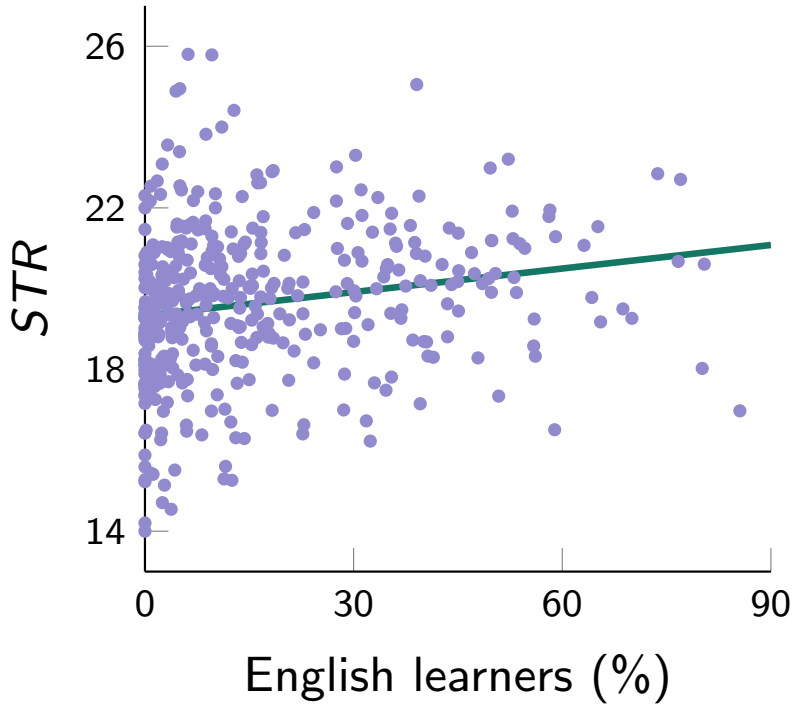
- Is its lower score caused entirely by class size?
- Or do language proficiency and other characteristics also matter?

Comparing class sizes also compares different types of districts, including both differences in STR and differences in PctEL in this case.

Source: Stock and Watson, `caschool.dta`; 420 California school districts.

The California data show both relationships

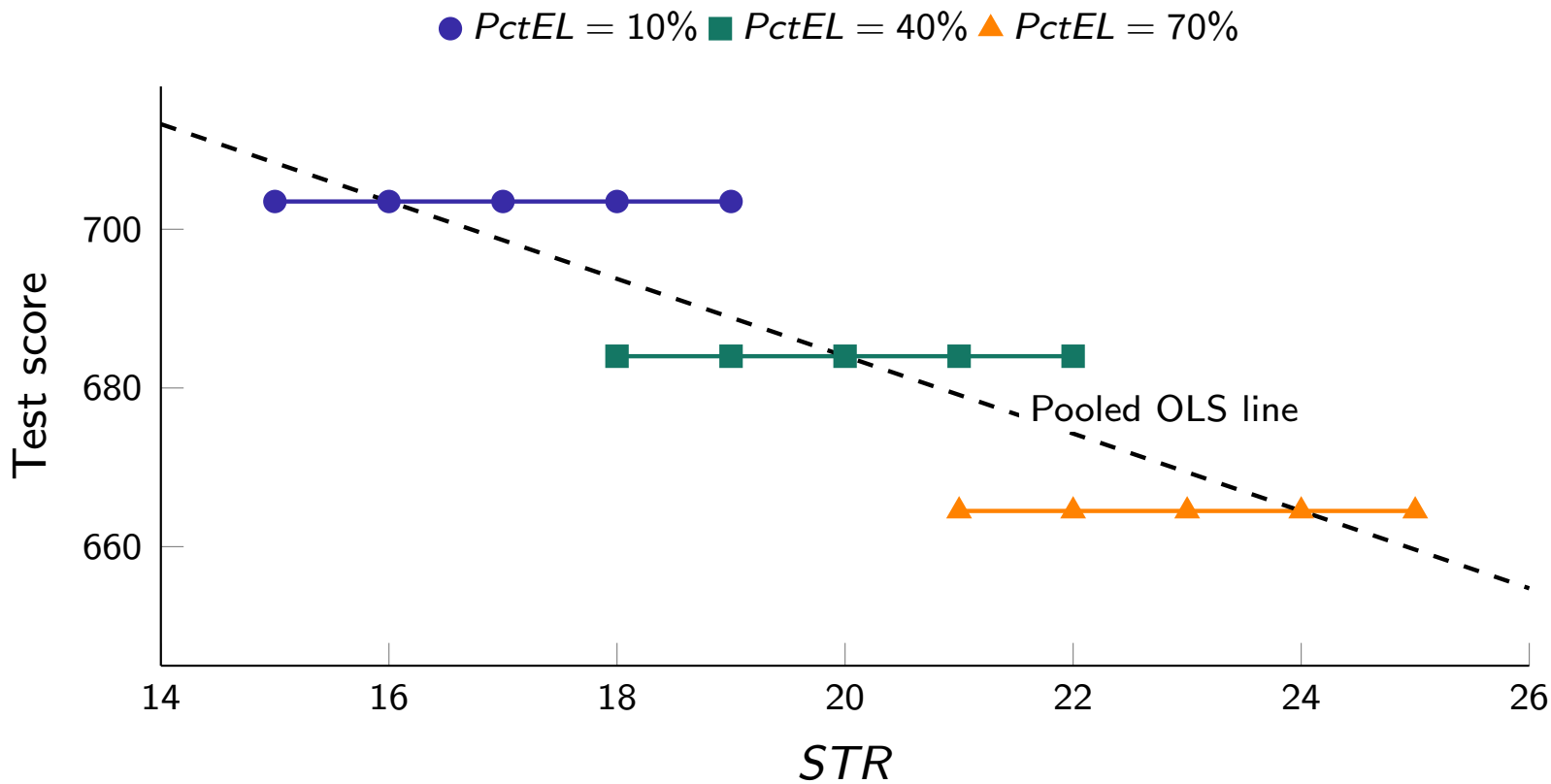




A higher English learner share is associated with lower scores and, more weakly, larger STR ($\text{Corr}(STR, PctEL) = 0.19$).

Source: Stock and Watson, `caschool.dta`; 420 California school districts.

A negative slope without a class size effect



Illustration, not the California data. Within each group with fixed PctEL, changing *STR* does not change scores; pooling the groups produces a negative slope.

The direction of omitted variable bias

Recall that OLS is distorted when X and the omitted factors in u move together. For large samples,

$$\widehat{\beta}_1 \xrightarrow{p} \beta_1 + \underbrace{\frac{\text{Cov}(X,u)}{\text{Var}(X)}}_{\text{omitted variable bias}}$$

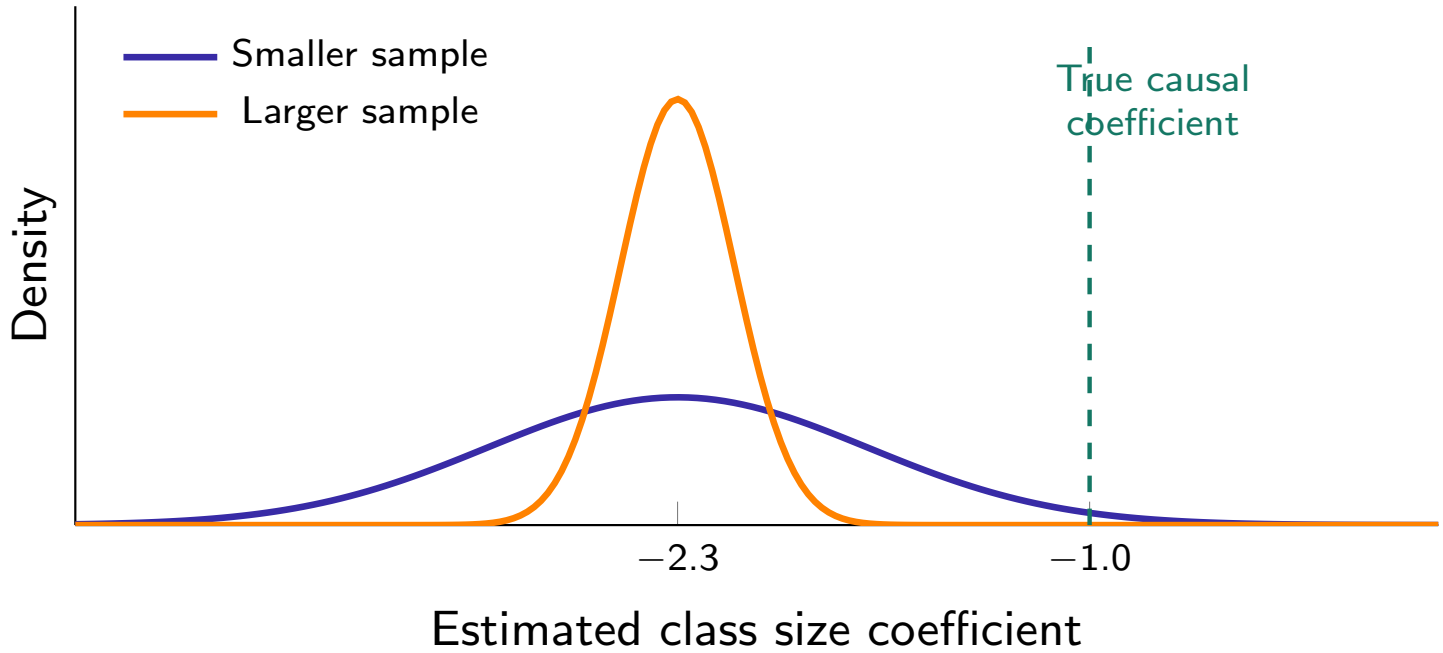
$$\frac{\text{Cov}(X,u)}{\text{Var}(X)} = \frac{\sigma_u}{\sigma_X}\text{Corr}(X,u).$$

Relationship	Bias direction	Estimated slope tends to be
$\text{Cov}(X,u) > 0$	Upward	Above β_1
$\text{Cov}(X,u) < 0$	Downward	Below β_1

In the English learner example: $STR \uparrow \implies PctEL \uparrow \implies u \downarrow$, this implies $\text{Cov}(STR,u) < 0$, so $\widehat{\beta}_1 < \beta_1$ in a large sample.

A negative class size effect would then look **more negative** than it is.

A larger sample does not remove bias



Schematic sampling distributions with persistent downward bias.

The estimates become more concentrated around the **wrong target**. A smaller standard error does not repair a failure of A1.

Why an ideal experiment helps

Suppose students and teachers are randomly assigned to class sizes, with full compliance and no other systematic changes across class assignments.

- English proficiency and teacher quality are **balanced on average** across different class sizes.
- Class size is unrelated to these omitted determinants of scores.
- In the causal model, this makes $E(u_i \mid STR_i) = 0$ plausible.

In observational data, families and schools decide class size. Small and large classes can therefore contain systematically different students.

Random assignment concerns how treatment is allocated. Random sampling concerns which districts enter the dataset. They solve different problems.

Comparisons with control variables

Instead of comparing all small-class districts with all large-class districts, compare districts with the **same percentage of English learners**.

The idea of a control variable

Hold an observed characteristic fixed so its differences do not get attributed to the regressor of interest.

We do this by including *PctEL* as another regressor:

$$\text{TestScore}_i = \beta_0 + \beta_1 STR_i + \beta_2 PctEL_i + u_i.$$

The error now contains the **remaining** omitted factors. The omitted variable effects from PctEL on STR and TestScore are controlled in the regression.

Section 6.2

The multiple regression model

Two regressors, two comparisons

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i.$$

For the school example, $X_{1i} = STR_i$ and $X_{2i} = PctEL_i$.

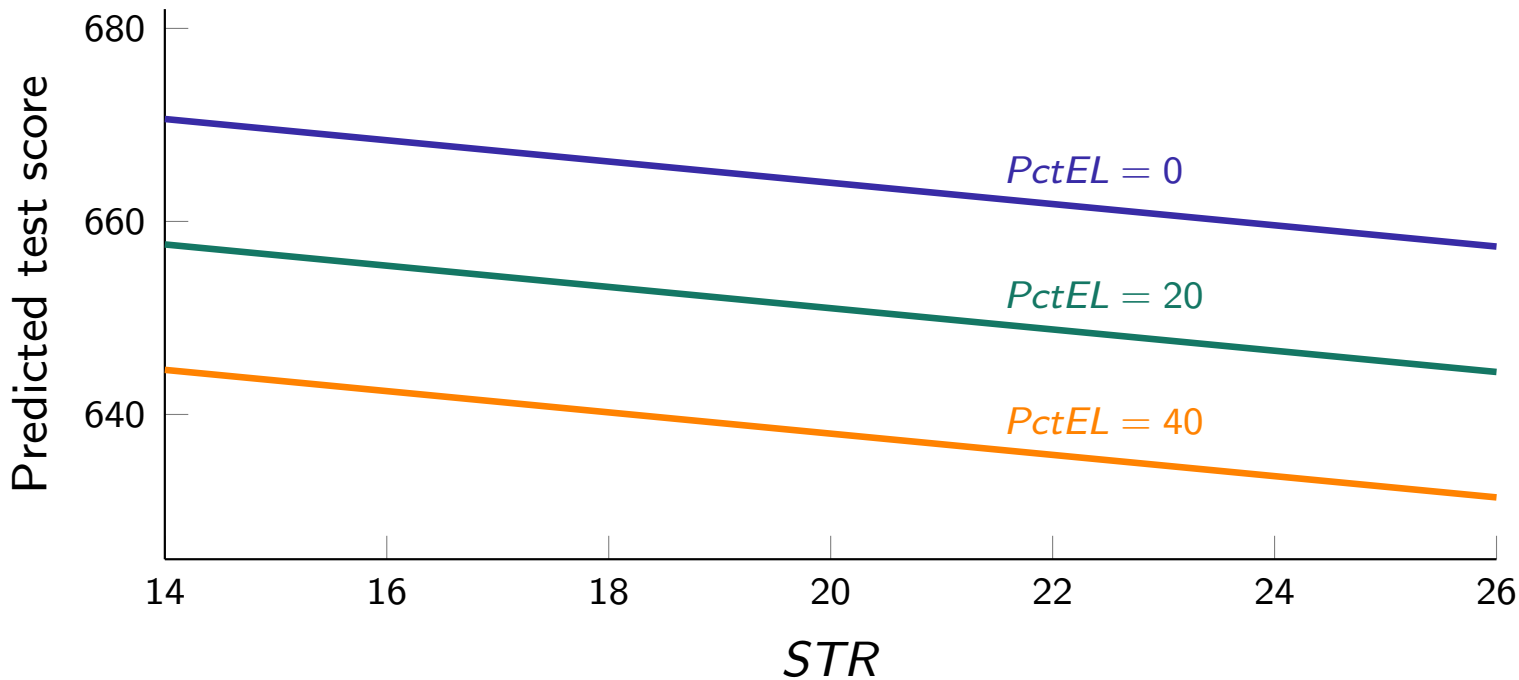
Coefficient	Meaning in the regression model
β_1	Change in predicted Y for a one-unit increase in X_1, **holding X_2 fixed**.
β_2	Change in predicted Y for a one-unit increase in X_2, **holding X_1 fixed**.
β_0	Predicted Y when both regressors equal zero.

A causal interpretation requires assumptions about the error. The equation alone does not supply them.

Holding the other regressor fixed

The fitted California regression, which we estimate next, is

$$\widehat{\text{TestScore}} = 686.03 - 1.10\,STR - 0.65\,PctEL.$$



Move along one line to change *STR* while keeping *PctEL* fixed. Each line has the same slope, −1.10, in this model.

Changes in more than one regressor

The fitted relationship implies

$$\Delta \widehat{Y} = \widehat{\beta}_1 \Delta X_1 + \widehat{\beta}_2 \Delta X_2.$$

- If X_2 is fixed, then $\Delta X_2 = 0$:

$$\Delta \widehat{Y} = \widehat{\beta}_1 \Delta X_1.$$

- If both variables change, include **both** contributions.

For the school model,

$$\Delta \widehat{\text{TestScore}} = -1.10\, \Delta STR - 0.65\, \Delta PctEL.$$

PctEL is measured from 0 to 100. A change from 10 to 20 is **10 percentage points**, so $\Delta PctEL = 10$.

The model with k regressors

The same idea extends to more observed characteristics:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + u_i, \qquad i = 1, \ldots, n.$$

- k regressors and $k + 1$ unknown coefficients, including the intercept.
- β_j describes a one-unit change in X_j, holding **all other included regressors** fixed.
- u_i contains what the included regressors leave unexplained.

For example, we might add district income or another measure of school resources.

Regressors are also called "independent variables." This name does **not** mean that they are statistically independent of one another.

Section 6.3

OLS in multiple regression

OLS uses the same criterion as before

For two regressors, OLS chooses the three coefficients that minimize **the sum of squared residuals**:

$$\min_{b_0, b_1, b_2} \sum_{i=1}^{n} \left[Y_i - (b_0 + b_1 X_{1i} + b_2 X_{2i}) \right]^2 .$$

- Stata searches for the best-fitting combination of coefficients.
- The resulting estimates are $\widehat{\beta}_0$, $\widehat{\beta}_1$, and $\widehat{\beta}_2$.
- With k regressors, OLS chooses $k + 1$ coefficients using the same criterion.

We use the same OLS principle from Handout 5 to estimate the multiple regression.

Adding a control changes the class size slope

Simple regression:

$$\widehat{\text{TestScore}} = 698.93 - 2.28\,STR.$$

Including the percentage of English learners:

$$\widehat{\text{TestScore}} = 686.03 - 1.10\,STR - 0.65\,PctEL.$$

- The class size slope moves from -2.28 to -1.10: it becomes less negative.
- This is consistent with the predicted direction of omitted variable bias.
- The controlled comparison associates one additional student per teacher with a 1.10-point lower score, holding *PctEL* fixed.

It is still possible that other omitted factors bias the controlled slope.

Source: Stock and Watson, `caschool.dta`; 420 California school districts.

Stata: add the control after the main regressor

Using the same dataset as Handouts 5 and 6:

```
* Simple regression
reg testscr str, robust

* Multiple regression
reg testscr str el_pct, robust
```

- `testscr`: test score; `str`: student–teacher ratio.
- `el_pct` is the dataset's name for *PctEL*.
- Continue using `robust` for heteroskedasticity-robust standard errors.

The `robust` option changes the standard errors, not the OLS coefficients. It does not correct omitted variable bias.

Reading the multiple regression output

Regressor	Coefficient	Robust SE	t	p-value
STR	-1.101	0.433	-2.54	0.011
PctEL	-0.650	0.031	-20.94	< 0.001
Intercept	686.032	8.728	78.60	< 0.001

- **Class size**: a one-unit increase in *STR* is associated with 1.10 fewer score points, holding *PctEL* fixed.
- **English learners**: one percentage point higher *PctEL* is associated with 0.65 fewer score points, holding *STR* fixed.
- **Intercept**: predicted score at $STR = PctEL = 0$. A district with no students per teacher is outside the useful range of this model.

Source: Stock and Watson, `caschool.dta`; 420 California school districts.

Fitted values and residuals

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 X_{1i} + \widehat{\beta}_2 X_{2i}, \qquad \widehat{u}_i = Y_i - \widehat{Y}_i.$$

Consider a hypothetical district with

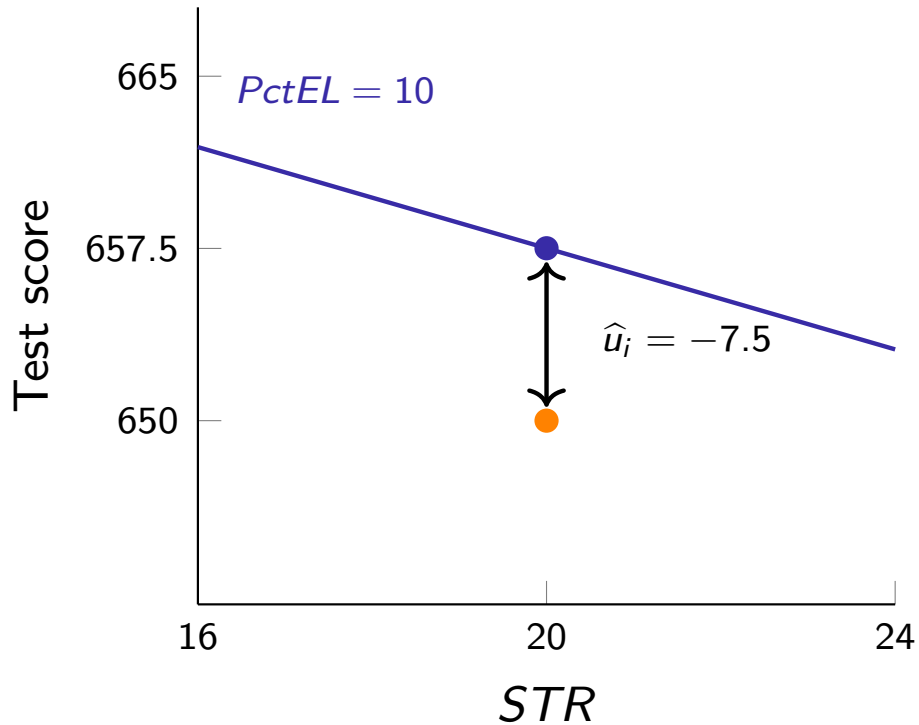
$$STR = 20, \quad PctEL = 10.$$

The fitted score is

$$\begin{aligned} \widehat{Y} &= 686.03 - 22 - 6.5 \\ &= 657.5. \end{aligned}$$

If its actual score is 650,

$$\widehat{u}_i = 650 - 657.5 = -7.5.$$



Section 6.4

Measures of fit

What measures of fit describe

As in Handout 5, define the residual sum of squares:

$$RSS = \sum_{i=1}^{n} \widehat{u}_i^2 = \sum_{i=1}^{n} (Y_i - \widehat{Y}_i)^2.$$

A smaller *RSS* means smaller squared prediction errors **in this sample**.

Measure	Question
SER	How far are observed outcomes typically from fitted outcomes, in units of Y?
R^2	What fraction of the sample variation in Y is explained?
Adjusted R^2	How does fit look after allowing for the number of regressors?

RSS is also called *SSR* in some texts. These measures describe sample fit, not how close estimated coefficients are to their population values.

SER and degrees of freedom

With k regressors and an intercept, we estimate $k+1$ coefficients:

$$\boxed{SER = \sqrt{\frac{RSS}{n-k-1}}.}$$

- One regressor: denominator $n-2$, just as in Handout 5.
- Two regressors: denominator $n-3$.
- *SER* is measured in test-score points in our school example.

The unadjusted root mean squared residual is $\sqrt{RSS/n}$.

Stata terminology: the number labeled `Root MSE` in `regress` uses $n-k-1$, so it is the *SER* defined here.

Why adding regressors cannot lower R^2

With an intercept,

$$R^2 = 1 - \frac{RSS}{TSS}, \qquad TSS = \sum_{i=1}^{n} (Y_i - \overline{Y})^2.$$

Compare two OLS regressions on the **same observations**, where the second includes all regressors in the first plus one more.

- The expanded model can always reproduce the old fit by setting its new coefficient to zero.
- OLS chooses a fit at least as good: RSS cannot rise.
- Therefore, R^2 can rise or stay the same; it cannot fall.

Even an unhelpful variable can improve sample fit by fitting chance variation. A higher R^2 alone is not a reason to include a variable.

Adjusted R^2

Adjusted R^2 compares residual variation and outcome variation using their degrees-of-freedom adjustments:

$$\boxed{\overline{R}^2 = 1 - \frac{RSS/(n-k-1)}{TSS/(n-1)} = 1 - \frac{n-1}{n-k-1}(1-R^2).}$$

- Adding a regressor reduces RSS, or leaves it unchanged, but also reduces the residual degrees of freedom $n-k-1$.
- Adjusted R^2 can therefore **rise or fall**.
- It is no larger than R^2 and can even be negative.

When n is large relative to k, the adjustment is small. Neither measure establishes a causal relationship.

Adding regressors: R^2 versus adjusted R^2

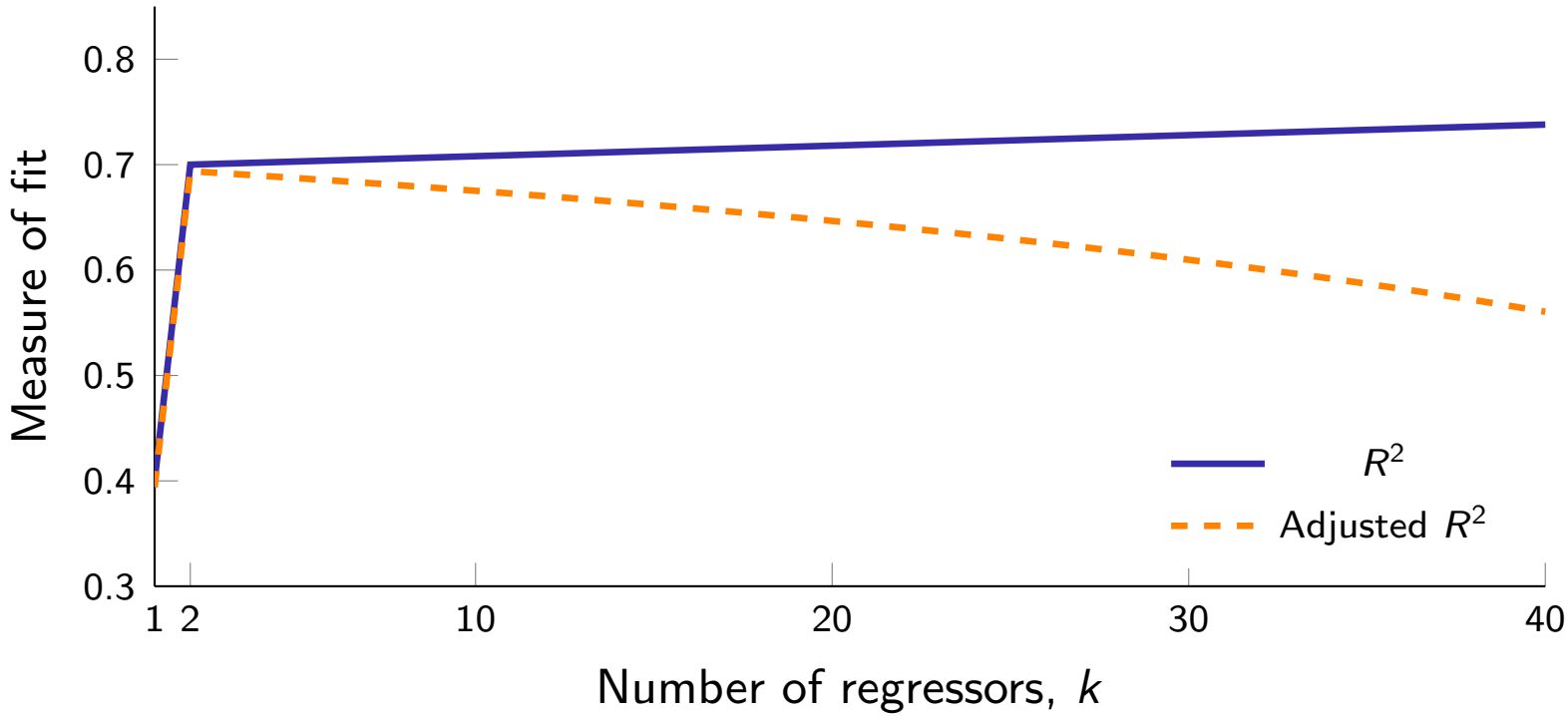
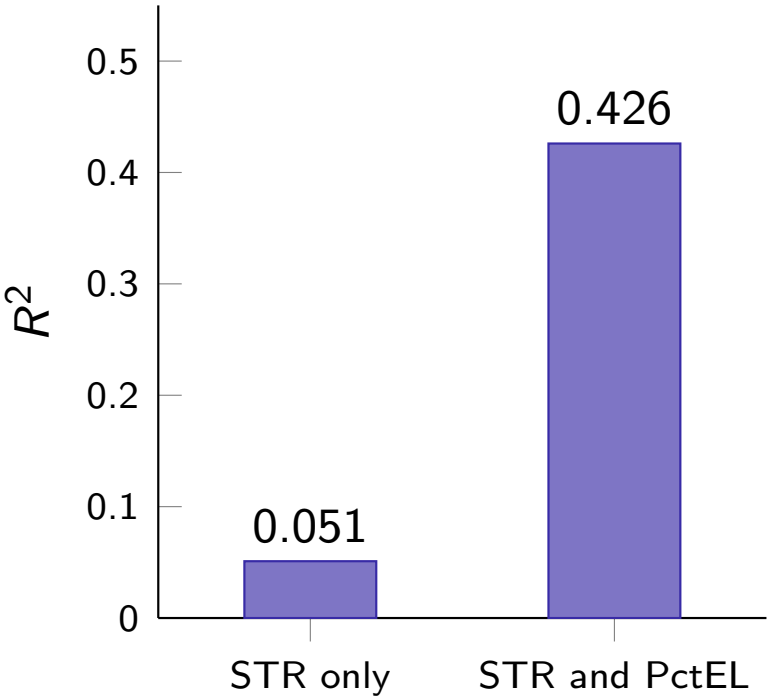


Illustration: $n = 100$, with one regressor added at each step on the same observations. The R^2 curve is illustrative; adjusted R^2 is calculated using its formula.

After the first two regressors, *RSS* falls only slightly. R^2 keeps rising, but the loss of degrees of freedom makes adjusted R^2 fall.

Fit improves after adding *PctEL*



	Simple	Multiple
k	1	2
R^2	.051	.426
$\overline{R}^2$	.049	.424
SER	18.58	14.46

The expanded model explains more sample variation and has smaller typical residuals. With $n = 420$ and only two regressors, R^2 and $\overline{R}^2$ are close.

Source: Stock and Watson, `caschool.dta`; 420 California school districts.

Choosing regressors for prediction or causality

Goal	What matters?
Prediction	Predict outcomes well for new observations. Sample fit is useful, but also check performance outside the estimation sample.
Causal inference	Make comparisons that isolate the effect of interest. Choose controls using knowledge of how the variables are determined.

A larger R^2 does not imply less omitted variable bias.

Section 6.5

Least squares assumptions in multiple regression

The four least squares assumptions

For $Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + u_i$:

	Assumption	Intuition
A1	$E(u_i \mid X_{1i}, \ldots, X_{ki}) = 0$	No systematic omitted-factor differences at the same regressor values.
A2	$(Y_i, X_{1i}, \ldots, X_{ki})$ are i.i.d.	Observations follow the same sampling rule and are independent across i.
A3	No outliers for Y_i and each regressor	Extremely large observations do not dominate the analysis.
A4	**No perfect multicollinearity**	No regressor is an exact linear combination of the others, including the constant.

A1–A3 extend Handout 5. A4 lets us estimate separate coefficients; we will visualize it in Section 6.7.

A1 with multiple regressors

For our two-regressor model, A1 becomes

$$E(u_i \mid STR_i, PctEL_i) = 0.$$

Among districts with the same included characteristics (STR and PctEL), the remaining omitted factors of test scores must average to zero.

- Adding *PctEL* accounts for observed differences in English learner share.
- But family income, teacher quality, or other omitted factors can still differ with class size after this adjustment.
- Adding a single control does **not automatically** make A1 true, researchers need to argue the remaining omitted factors do not correlate with the included regressors.

This assumption interpret **all coefficients causally**. Section 6.8 instead focuses on the class size effect alone with a weaker assumption.

Section 6.6

Sampling uncertainty in multiple regression

Sampling distributions in multiple regression

Take another random sample and re-estimate the same model. The OLS coefficients will generally change.

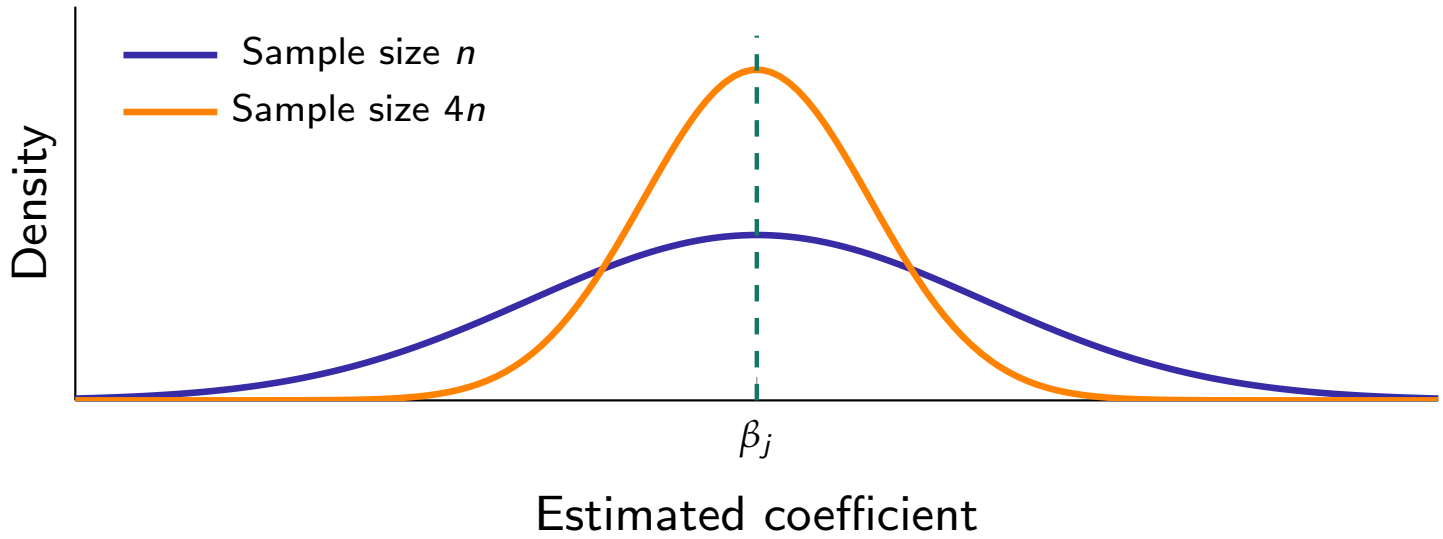
Under A1–A4, with enough independent variation to estimate each coefficient:

- OLS is unbiased: $E(\widehat{\beta}_j) = \beta_j$.
- OLS is consistent: $\widehat{\beta}_j \xrightarrow{p} \beta_j$.
- In large samples, each standardized coefficient is approximately normal:

$$\frac{\widehat{\beta}_j - \beta_j}{\mathrm{SE}(\widehat{\beta}_j)} \overset{\text{approx.}}{\sim} N(0,1).$$

We use robust standard errors. We skip the variance derivation and directly interpret the Stata output.

Sample size and precision



Schematic large-sample distributions when the assumptions hold and the underlying population and regressor set stay the same.

Variance is approximately proportional to $1/n$; standard errors to $1/\sqrt{n}$. Quadrupling n roughly halves the standard error.

Smaller error variance also reduces slope variance, holding the regressors fixed. With multiple regressors, precision depends on their **joint variation**, so the role of regressor variance is more complicated than in simple regression.

Confidence intervals and tests

For any coefficient β_j, a large-sample 95% interval is

$$\widehat{\beta}_j \pm 1.96\ \mathrm{SE}(\widehat{\beta}_j).$$

For $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$, use the same three rules:

- zero lies outside the 95% confidence interval;
- $|t| = |\widehat{\beta}_j / \mathrm{SE}(\widehat{\beta}_j)| > 1.96$;
- the two-sided p-value is below 0.05.

In the controlled class size regression:

$$\widehat{\beta}_1 = -1.101, \quad \mathrm{SE} = 0.433, \quad t = -2.54, \quad 95\%\ CI \approx [-1.95, -0.25].$$

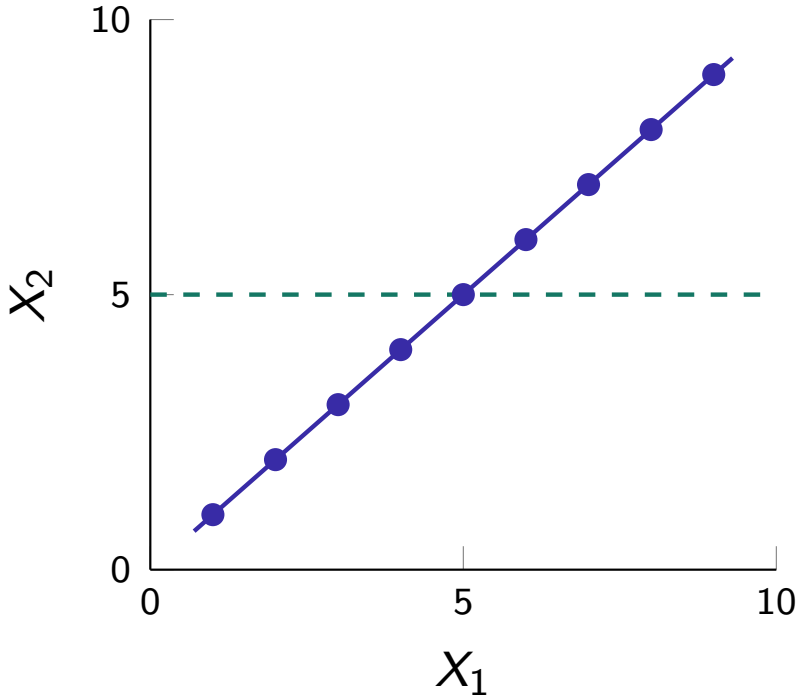
Stata reports $p = 0.011$: significant at 5%, but not 1%.

Section 6.7

Multicollinearity

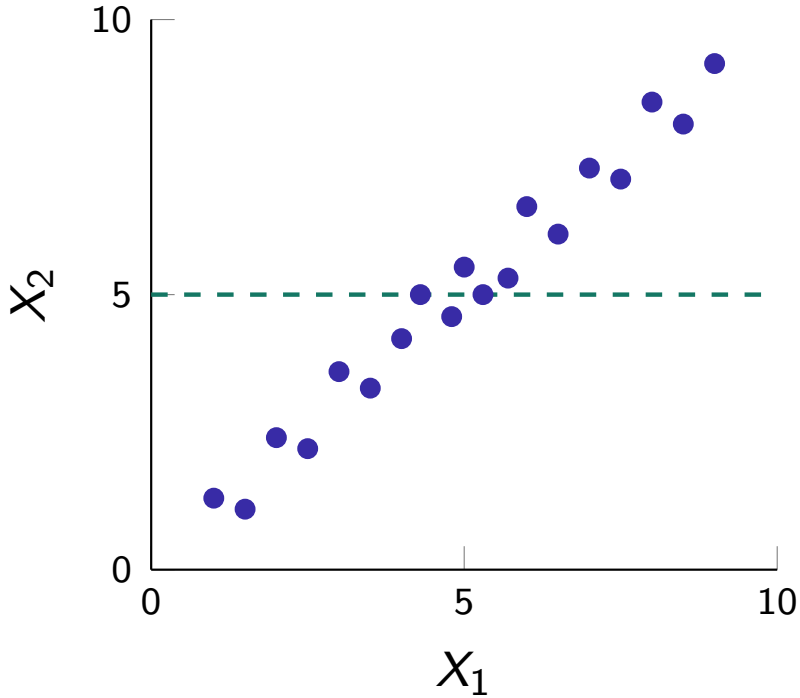
Separate variation in the regressors

Perfect multicollinearity



At $X_2 = 5$, X_1 must equal 5. Separate changes are impossible.

Imperfect multicollinearity



Near $X_2 = 5$, there is some variation in X_1, but very little.

Illustrative data. The issue is how much **separate variation** the data contain in each regressor, not simply how many regressors we include.

Perfect multicollinearity: duplicate regressors

Suppose $STRcopy_i = STR_i$ for every district. Then

$$Y_i = \beta_0 + \beta_1 STR_i + \beta_2 STRcopy_i + u_i = \beta_0 + (\beta_1 + \beta_2) STR_i + u_i.$$

Only the sum $\beta_1 + \beta_2$ can be learned from the data.

```
generate str_copy = str
reg testscr str str_copy, robust
```

Stata omits one of the redundant regressors because of collinearity.

Fix: remove the duplicate. The full model has no unique set of OLS coefficients; the reduced model can be estimated.

The dummy variable trap

Divide districts into two exhaustive groups:

$$Small_i = 1\{STR_i \leq 20\}, \qquad Large_i = 1\{STR_i > 20\}.$$

Every district belongs to exactly one group, so

$$Small_i + Large_i = 1.$$

District group	Constant	*Small*	*Large*
$STR \leq 20$	1	1	0
$STR > 20$	1	0	1

Including the constant and both indicators creates perfect multicollinearity: the constant is exactly the sum of the two indicators.

The same problem arises with an intercept and all three indicators for *Rural*, *Suburban*, and *Urban*.

Avoiding the dummy variable trap

```
generate large = (str > 20) if !missing(str)
generate small = (str <= 20) if !missing(str)

* Solution 1: intercept and one indicator
reg testscr large, robust

* Solution 2: both indicators, no intercept
reg testscr small large, noconstant vce(robust)
```

- **Solution 1**: the intercept is the small-class group mean; the coefficient on `large` is the large-minus-small difference.
- **Solution 2**: the two coefficients are the two group means.

With m mutually exclusive, exhaustive categories, include an intercept and $m - 1$ indicators. The omitted category is the reference group.

Imperfect multicollinearity and precision

Suppose *PctEL* is highly predictable from another included characteristic, such as the percentage of residents who are recent immigrants.

- To estimate the *PctEL* coefficient, we must compare districts with different *PctEL* but the **same values of the other regressors**.
- If the regressors almost always move together, such comparisons contain little variation.
- OLS can still be computed, but the affected coefficients have large sampling uncertainty and large standard errors.

High correlation alone does **not** create bias when the model's assumptions hold. Dropping an important control can instead create omitted variable bias.

Perfect versus imperfect multicollinearity

	Perfect	Imperfect
Relationship	Exact linear dependence among regressors, including the constant	Strong but inexact linear relationship
OLS coefficients	Cannot uniquely estimate all of them	Can estimate them if there is no exact dependence
Main problem	Cannot separate the coefficients at all	Can separate them only imprecisely
Response	Remove redundancy or change the coding	Recognize uncertainty; seek more independent variation

Perfect multicollinearity can involve **several** regressors even when no pair has correlation exactly $+1$ or -1.

Section 6.8

Control variables and causality

Treatment and control variables

The superintendent wants the effect of **class size**, rather than the causal effects of every district characteristic.

Role	School example	Purpose
Treatment X	*STR*	The variable whose causal effect is the research question.
Control W	*PctEL*	An observed characteristic used to make the treatment comparison more credible.

Both enter the regression:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + u_i.$$

Being a regressor does not make a variable randomly assigned. The treatment coefficient and control coefficient need not have the same causal status.

A weaker assumption for the effect of X

Suppose we only want the causal slope on X, rather than causal effects for every regressor.

- **Earlier A1** requires
$$E(u_i \mid X_i, W_i) = 0.$$
- **Conditional mean independence** only requires
$$E(u_i \mid X_i, W_i) = E(u_i \mid W_i).$$

The average omitted factors may vary with W, but not with X once W is held fixed. This is why the condition is **weaker than A1**.

With A2–A4 and appropriately specified controls, this condition can **replace A1 for estimating the causal effect of** X. It does not make the coefficient on W causal.

Comparable districts after controlling

Compare districts with the **same** *PctEL*.

What we need for the class size comparison

Differences in *STR* within these comparisons must be unrelated to remaining omitted determinants of scores, on average.

This is the idea of **conditional mean independence**:

$$\boxed{E(u_i \mid X_i, W_i) = E(u_i \mid W_i).}$$

Once W is held fixed, knowing X does not change the average omitted factors.

For example, if family resources still differ systematically with *STR* among districts with the same *PctEL*, the comparison is still confounded.

Interpreting treatment and control coefficients

With suitable controls, a credible identifying assumption, and an appropriate regression specification:

- the *STR* coefficient can estimate the causal effect of class size;
- the *PctEL* coefficient need NOT estimate the causal effect of changing the English learner share.

The control can still reflect differences on family background or neighborhood resources.

Its role is to make **class size** as-if randomly assigned when controlling for these observed differences.

New notions and methods in Handout 7

1. **Omitted variable bias**: an omitted determinant of Y that moves with X can shift the slope; a larger sample does not fix it.
2. **Multiple regression**: interpret each coefficient holding other included regressors fixed.
3. **OLS and uncertainty**: fitted values, residuals, robust SEs, confidence intervals, and tests follow the familiar logic.
4. **Fit**: R^2 cannot fall when regressors are added on the same sample; adjusted R^2 can fall.
5. **Multicollinearity**: exact dependence prevents separate estimation; strong inexact dependence reduces precision.
6. **Causal controls**: credible comparisons and an appropriate model matter more than a high R^2 or a small p-value.